HARMILAP SINGH DHALIWAL

AI Engineering Lead · Agentic AI & Enterprise Systems

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AI engineering lead with 8+ years building applied ML and production AI systems. Conceived, architected and built the bank's first true agentic AI platform approved for production, launched across the full CFO Group in May 2026. Hands-on across orchestration, semantic retrieval, human review and evaluation, with technical direction, team development and cross-functional delivery.

PROFESSIONAL EXPERIENCE

AI & Data Science Lead, Finance | Royal Bank of Canada

April 2025 - Present

CFO Group, Enterprise Finance · Toronto, ON

Remain ~70% hands-on across architecture, modeling, full-stack implementation, evaluation, and production operations while leading cross-functional delivery.

Production AI Systems

- Built and launched the bank's first production-approved agentic AI platform, owning the core application stack across ETL, document ingestion, PostgreSQL/pgvector dense retrieval, a reviewable authoring workflow, and a single-agent LangGraph orchestrator with bounded MCP tool routines. Piloted in April 2026; launched across the full CFO Group in all geographies in May 2026.
- Conceived, architected, and built the AI/LLM Workforce Analytics Platform from March through its November 2025 production launch, leading cross-functional delivery. It supports authorized headcount, compensation-cost, and open-position questions across ~40,000 cost centres; model stages route intent while deterministic Python retrieves and calculates within the resolved entitlement scope.
- Led a focused two-week v1-to-v2 refactor of the production Financial Peer Benchmarking Platform: dense semantic retrieval ranks candidate metrics; LLM-assisted disambiguation resolves near-duplicate names; reviewed templates and parameter binding constrain text-to-SQL execution.

Technical Leadership

- Set architecture, code-review, and refactoring standards for retrieval, model behavior, and SQL guardrails; manage one Senior AI Scientist, guided the 2025 Amplify cohort, and helped junior contributors ship production changes.
- Own application evaluation and post-release follow-through: prepare LLM-as-judge evidence and extensive human-test results for Model Risk review, backed by custom tracing and logging.

Enterprise Finance Senior Data Scientist | Royal Bank of Canada

September 2022 - April 2025

CFO Group, Enterprise Finance · Toronto, ON

- Built v1 of the Financial Peer Benchmarking Platform end to end, automating text extraction and entity matching across quarterly-changing disclosures from Canada's six largest banks; removed a long-standing peer-analysis bottleneck and earned the 2025 CFO One RBC Team Award.
- Cut commodity-tax return preparation from months to 90 minutes by building a PySpark pipeline supporting ~$600M in allocations from enterprise general-ledger data; earned the CFO Group RBC Quarterly Team Award (Q4 2023).

EARLIER EXPERIENCE

Machine Learning Engineer | Quantiphi Inc.

October 2021 - September 2022

Toronto, ON

- Improved checkbox-detection accuracy from ~70% to 99.95% on Humana claims documents by layering OpenCV pixel-level detection and Random Forest classification over OCR.
- Built document-classification and entity-extraction workflows and deployed custom and AutoML models on Vertex AI; developed supporting data infrastructure with Bigtable and BigQuery.

Data Scientist | Tata Consultancy Services

August 2016 - November 2019

Pune, India

- Delivered $3M in annual savings for a 900 MW generating unit at Kansai Electric's 1,800 MW Maizuru coal-fired station in Japan. As sole data scientist on a three-person R&D team, built the combustion-tuning digital twin end-to-end: ETL across 90+ sensors, 84 regression models, and Particle Swarm Optimization that produced operator-reviewed boiler-tuning recommendations.
- Designed the model-selection framework using k-fold cross-validation, R-squared, RMSE, MAPE, and fold variance; also shipped LSTM forecasting and XGBoost transformer-life prediction models.
- Placed 2nd of 600 in TCS's computer-vision hackathon with a CNN math-notation detector; earned the Innovation Pride Award (2019).

TECHNICAL SKILLS

Programming & Production	Python, SQL, PySpark, ETL, API service design, PostgreSQL/pgvector, Git/GitHub, observability; production delivery through partner-owned OpenShift CI/CD
Machine Learning & Evaluation	PyTorch, TensorFlow, scikit-learn, XGBoost; regression, classification, clustering, time series, CNNs, Transformers; model selection and evaluation
Retrieval & LLM Systems	Document classification and extraction, named entities, embeddings, dense retrieval, semantic ranking, near-duplicate disambiguation, dense RAG, guarded text-to-SQL, LangGraph, MCP, human-in-the-loop workflows, LLM evaluation
Cloud & Data	Google Cloud Platform, Vertex AI, AutoML, BigQuery, Bigtable

EDUCATION

Post Graduate Professional Certificate, Big Data Analytics — Georgian College · 2021

B.Eng., Electronics & Communications Engineering — Thapar University · 2016

AWARDS & RECOGNITION

CFO Conference Award | RBC — 2026

CFO Individual Quarterly Award | RBC — P4 2026

CFO One RBC Team Award | RBC — 2025

CFO Group RBC Quarterly Team Award | RBC — Q4 2023